Bryam Steve Pinilla Jurado

PhD(c) in Astronomy

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Education

March 2023–December 2026 **Ph.D. in Sciences, Astronomy**, *Universidad de Chile*, Santiago, Chile
Thesis: *Combining Theory and Multimodal Observations to Classify and Characterize Supernovae*, under the supervision of **Dr. Francisco Förster Burón**. Qualifying examination passed in April 2024; thesis project approved in May 2024. All mandatory coursework completed (Radiative Processes, Observational Astronomy, Galactic Astronomy, Galaxies, Research Workshops I–IV), plus elective courses in Machine Learning and Astroinformatics.

March 2014–July 2022 **B.Sc. in Astronomy**, *Universidad de Antioquia*, Medellín, Colombia, Graduating rank: 1/10
Thesis: *Spectral characterization of the exoplanet host stars HAT-P-11 and HD 179949 with the TIGRE telescope*, under the supervision of **Dr. Lauren Flor-Torres** and **Dr. Camilo Delgado-Correal**.

Research Interests

- **Astroinformatics & Astrostatistics.** Machine learning and deep learning applied to astronomical data; multimodal generative models; classification and characterization of transient events; large-scale data analysis.
- **Stellar Astrophysics.** Stellar evolution, stellar spectroscopy, supernovae and transient phenomena.
- **Astrochemistry.** Computational and observational searches for molecular species in circumstellar environments.
- **Physics-Informed Neural Networks (PINNs).**

Research Experience

2024–Present **Doctoral research — Multimodal classification of transients**, *Universidad de Chile*, Santiago, Chile
Extension of a variational autoencoder architecture (ParSNIP, Boone 2021) to incorporate multimodality, a novel feature in this research area. Construction of the full data pipeline: ingestion of light curves from the **AleRCE** broker and of spectra from the **WISeREP** database, with preprocessing, smoothing and filtering of irregular time series. Implemented in PyTorch on HPC infrastructure. Developed in consultation with **Prof. Andrew Connolly** (University of Washington; Director of the eScience Institute and of DiRAC).

2023–Present **Astrochemistry — Sulfur-bearing molecules in IRC+10216**, *Universidad de Chile*, Santiago, Chile

Computational and observational search for sulfur-bearing molecules (HCSCN, HCSCCH, c-C_2H_2S) around the carbon star IRC+10216, using data from the APEX telescope.

2021–2022 **RECA Research Internship**, *International collaboration*, Remote

Statistical characterization of the transition phase in the light curves of hydrogen-rich supernovae, to constrain their progenitors and explosion properties. Parameter estimation carried out within a **Bayesian framework** using **Markov Chain Monte Carlo (MCMC)** sampling, including posterior inference and uncertainty quantification. Supervised by **Dr. Claudia P. Gutiérrez** (Universitat de Barcelona) and **Dr. Santiago González-Gaitán** (CENTRA, Universidade de Lisboa).

2022–2024 **Junior Researcher**, *Fundación Universidad del Valle — Macondo Astronomy & Astrophysics Group (GAAM)*, Cali, Colombia

Collaboration with Dr. Lauren Flor-Torres and Dr. Camilo Delgado-Correal.

2021 **Project Leader — 100 Hours 100 Schools**, *IASC / Las Cumbres Observatory*, Remote

Coordination of allocated observing time (90-minute blocks yielding 7–9 images per target) and delivery of analysis sessions on deep-sky objects, planets and minor bodies with school students.

Publications

- **Jurado, S.**, Gutiérrez, C. P. & González-Gaitán, S., "Characterization of the Transition Phase in the Light Curves of Hydrogen-Rich Supernovae", *in preparation.*
- **Jurado, S.** & Förster, F., "Multimodal Physical Variational Autoencoder for Supernova Light Curves and Spectra", *in preparation.*

Conference Presentations

- **IV Chilean Astrochemistry School** (2025) — Talk: *Computational and Observational Search for Sulfur-Bearing Molecules using the APEX telescope.* Also served on the organizing (logistics) committee.
- **First LSST Latin America Meeting** (2024), Universidad de La Serena, La Serena, Chile — Poster: *Combining Theory and Multimodal Observations to Classify and Characterize Transient Events.*
- **1st Colombian Symposium on Astrochemistry** (2024) — Talk: *Computational and Observational Search for Sulfur-Bearing Molecules in the carbon star IRC+10216 using the APEX telescope.*
- **Workshop in Computational Chemistry and Molecular Spectroscopy** (2023), Concón, Chile — Poster: *Computational and Observational Search for HCSCN, HCSCCH and c-C_2H_2S Sulfur Molecules around the carbon star IRC+10216.*
- **Colombian Meeting of Astronomy & Astrophysics (COCOA)** (2022), Tunja, Colombia — Talk.

- **Correcting Reddening Intelligently for Supernova Probes (CRISP)** (2021), virtual — Talk: *Characterization of the Light-Curve Transition of Hydrogen-Rich Supernovae.* `https://sn-crisp.github.io/CRISP2021/`
- **SuperVirtual: A Virtual Conference on Supernovae and Related Transients** (2021), virtual — Poster: *Characterization of the Light-Curve Transition of Hydrogen-Rich Supernovae.* `https://sites.google.com/view/supervirtual2021`

Software & Open-Source Projects

2025 **SNFit — Python package for supernova light-curve fitting**, *CodeAstro*
`pip`-installable package for polynomial fitting of supernova light curves and distance estimation from peak luminosity. Collaborative development with version control, testing and packaging. `https://github.com/moira-andrews/SNFit`

2025 **TrueStarVrotVelocity — Stellar rotation velocity estimation**
Analysis of rotational velocities in stellar clusters using probability density estimation methods. `https://github.com/jurados/TrueStarVrotVelocity`

Teaching Experience

2026 **Teaching Assistant — AI Agents**, *Chilean Financial Market Commission (CMF)*, Santiago, Chile
Design and delivery of a training programme (4 h/week) for CMF teams on the development of LLM-based agents and subagents, including tool integration via the Model Context Protocol (MCP), applied to the automation of analysis, document extraction and evaluation tasks.

2025–2026 **Teaching Assistant — Astroinformatics**, *Universidad de Chile*, Santiago, Chile
Weekly sessions (2 h/week) on machine learning and deep learning applied to real astronomical data, for undergraduate astronomy students and graduate students from several disciplines. Developed the complete practical material in Python (notebooks, exercises and assessments).

2024 **Teaching Assistant — Diploma Programme in Astronomy**, *Department of Astronomy (DAS), Universidad de Chile, and National Astronomical Observatory (OAN)*, Santiago, Chile
Academic support to students in the first edition of the programme.

2024 **Research Mentor — Explora RM Sur Poniente**, *Colegio San Ignacio*, Santiago, Chile
Mentored two secondary-school student teams in their research projects: a study of Thorne–Żytkow objects, and the design of a Galileoscope built from recycled materials available in a public institution.

2016–2022 **Teaching Assistant I/II**, *Universidad de Antioquia*, Medellín, Colombia
Basic Mathematics Seedbed programme: 4 h/week of mathematics teaching for 8th–11th grade students and secondary-school graduates.

Outreach & Science Communication

2024 **Science outreach**, *Center for Astrophysics and Associated Technologies (CATA)*, Santiago Centro, Chile
Public astronomy outreach activities in the Santiago Centro district.

2018–2020 **Science Communicator and Logistics Assistant**, *Inmerdome*, Medellín, Colombia
Production and presentation of science shows in a portable astronomical dome (mobile planetarium), and delivery of astronomy workshops for audiences of all ages.

Summer Schools, Conferences & Workshops

2025 **Machine Learning and Astrostatistics School: Applications to Massive Stars**, Valparaíso, Chile

2025 **CodeAstro: A Software Engineering Workshop for Astronomy**, Evanston, Illinois, USA
Version control, testing and packaging of scientific software. `https://semaphorep.github.io/codeastro/`

2024 **Latin American Conference on High Performance Computing (CARLA)**

2024 **Introductory and Advanced Courses on the Leftraru EPU Cluster**, *National Laboratory for High Performance Computing (NLHPC)*, Santiago, Chile

2024 **Summer School in Statistics for Astronomers**, *Center for Astrostatistics, Pennsylvania State University*, Pennsylvania, USA

2024 **1st UNAB Astrochemistry Workshop**, *Universidad Andrés Bello*, Santiago, Chile

2022 **International Virtual Courses (IVC): Introduction to Astrophysical Modeling and Computation**, *Institut Teknologi Bandung*, Indonesia

2022 **Astronomy Python Bootcamp**, *Red de Estudiantes Colombianos de Astronomía*

2022 **1st Latin American Gaia School**, *Universidad de Antioquia*, Medellín, Colombia

2021 **Summer School in Astrophysics (EVA 2021)**, *Institute of Radio Astronomy and Astrophysics (IRyA), UNAM*, Virtual, Mexico

2019 **Colombian Meeting of Astronomy & Astrophysics (COCOA)**, Medellín, Colombia, Attendee

Complementary Education

2025 **Leadership and Team Development**, *Universidad de Chile*, Santiago, Chile

2022 **Data Science for All (DS4A)**, *MinTIC and Correlation One*, Colombia, 375 hours
Final project: predictive model of the efficiency of a coffee-drying boiler from physical process variables, for Café Buendía.

2020 **High-Dispersion Spectroscopy for Astronomy and the Large-Scale Structure of the Universe (AstroTwinColo)**, *Universidad de Antioquia*, Medellín, Colombia

2019 **Machine Learning in Astronomy and Gravitational Lensing (AstroTwinColo)**, *Universidad de Antioquia*, Medellín, Colombia

2017 **Methodological Tools in Research: Processes in Science, Technology and Research**, *SENA*, Cundinamarca, Colombia, 40 hours

2016 **Radio Astronomy with ALMA and Introduction to Astrochemistry (AstroTwinColo)**, *Universidad de Antioquia*, Medellín, Colombia

2013 **INGENIO Research Incubator Programme**, *Universidad Autónoma de Manizales*, Manizales, Colombia, 40 hours

2012 **Comprehensive Leadership Programme**, *IPlay*, Manizales, Colombia, 120 hours

2010 **English Language Proficiency**, *Centro de Idiomas Americanos (CIA)*, Manizales, Colombia, 120 hours

Awards & Honours

- **ANID National Doctoral Scholarship** (2023–2026), Chile.
- **Academic Honour — Best student of the Physics Institute**, Universidad de Antioquia, Medellín, Colombia (2014, 2015 and 2020).
- **Graduating rank 1/10**, B.Sc. in Astronomy, Universidad de Antioquia.

Technical Skills

Languages	Python (advanced), SQL, C, Bash
ML / DL	PyTorch, scikit-learn, TensorFlow, Weights & Biases
Statistics	Bayesian inference, Markov Chain Monte Carlo (MCMC), emcee, density estimation, uncertainty quantification
LLMs / Agents	LLM APIs, Model Context Protocol (MCP), agent and subagent design
Data	NumPy, pandas, SciPy, Astropy, Matplotlib, Jupyter
Astronomy software	iSpec, IRAF
Infrastructure	Git/GitHub, Linux, HPC (SLURM), `pip` packaging, LaTeX

Languages

Spanish	Native
English	B2, certified by Global World